

Performance Evaluation of xApp-based PRB Allocation in an O-RAN IAB Testbed

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Abstract—This paper presents an experimental study of Physical Resource Block (PRB) allocation in a multi-hop Open Radio Access Network (O-RAN) Integrated Access and Backhaul (IAB) system. To address the high deployment cost of fiber-based backhaul and the implementation complexity of relay protocols, we build a practical IAB testbed using OpenAirInterface (OAI), Universal Software Radio Peripherals (USRPs), and Ubuntu hosts with low-latency kernels. Instead of relying on the full 3GPP Backhaul Adaptation Protocol (BAP), the proposed platform adopts a lightweight BAP-less design based on Linux IP forwarding and static routing. On top of this platform, we integrate a near-Real-Time Radio Intelligent Controller (near-RT RIC) and implement an xApp that adjusts the PRB split between backhaul and access links. We then compare fixed allocation ratios, such as 50/50, 30/70, and 10/90, together with an adaptive policy and the default proportional fair (PF) scheduler, and analyze their impact on end-to-end throughput and latency under realistic SDR constraints. The results show that different PRB allocation methods lead to different throughput-latency trade-offs across network conditions, and that the proposed platform is suitable for experimentally characterizing these scheduling behaviors rather than claiming that xApp-based control universally outperforms either fixed-ratio allocation or PF scheduling.

Index Terms—O-RAN, Integrated Access and Backhaul (IAB), PRB Allocation, xApp, near-RT RIC, OpenAirInterface, Testbed

I. INTRODUCTION

The rapid densification of 5G networks has intensified the need to expand coverage and capacity without incurring a proportional increase in deployment cost. In practice, provisioning fiber backhaul for every small cell remains both costly and time-consuming, particularly in dense urban areas, temporary venues, and geographically constrained environments. Integrated Access and Backhaul (IAB) has emerged as a promising solution to this challenge by enabling the same radio infrastructure and spectrum resources to support both user access and wireless backhaul, thereby reducing fiber dependence while preserving deployment flexibility [1], [2].

At the same time, the Open Radio Access Network (O-RAN) paradigm has introduced a more open and programmable framework for radio control. By disaggregating hardware and software and standardizing key interfaces, O-RAN allows operators to deploy intelligent control functions across multivendor platforms. In particular, the Radio Intelligent Controller (RIC) provides a practical mechanism for

monitoring network conditions and enforcing near-real-time optimization through xApps [3]–[5].

However, the combination of multi-hop IAB and O-RAN also creates a challenging resource-allocation problem. In relay-based topologies, Physical Resource Blocks (PRBs) must be partitioned between backhaul and access links. When traffic demand varies across hops, a fixed or poorly matched PRB split can lead to persistent bottlenecks, queue buildup at intermediate nodes, and degraded end-to-end Quality of Service (QoS). This issue becomes even more pronounced in real SDR deployments, where scheduling latency, software processing overhead, and implementation-specific queue dynamics all affect measured performance.

In this paper, we address the gap between simulation-oriented IAB research and practical O-RAN implementation. We develop a multi-hop O-RAN IAB testbed based on OpenAirInterface (OAI), replace the heavyweight Backhaul Adaptation Protocol (BAP) data path with native Linux routing, and deploy an xApp on a near-RT RIC to control PRB allocation in real time. Rather than arguing that a customized policy should always outperform the default proportional fair (PF) scheduler, we use this platform to compare multiple PRB allocation methods and to characterize how different allocation strategies shape bottlenecks, throughput, and latency in a physical test environment.

The main contributions of this paper are summarized as follows:

- We build a real multi-hop O-RAN IAB testbed using OAI, USRP hardware, and Ubuntu PREEMPT_RT hosts, and we demonstrate a practical BAP-less forwarding architecture for SDR-based relay deployment.
- We design and integrate a custom xApp on a near-RT RIC to configure and evaluate multiple MAC-layer PRB allocation methods between backhaul and access links, including fixed-ratio and adaptive policies.
- We validate the proposed architecture with experimental measurements and show how different PRB splits lead to different throughput-latency trade-offs, which can then be compared against the default proportional fair (PF) scheduler.

II. RELATED WORK

Integrated Access and Backhaul has been extensively studied as a cost-effective approach to extending 5G coverage without requiring fiber connectivity at every small cell. Early studies primarily focused on the architectural feasibility, protocol implications, and deployment potential of IAB, particularly in dense mmWave settings where access and backhaul must share spectrum efficiently [1], [2]. These foundational works established the importance of jointly considering access and backhaul resources, but they mainly emphasized system architecture, standardization, and simulation-based performance characterization.

Subsequent research has examined resource allocation and routing in IAB networks in greater detail. Representative studies include semi-centralized resource management for 5G NR IAB and decentralized joint resource allocation and path selection for multi-hop topologies [6], [7]. In this line of work, the balance between backhaul and access resources is typically formulated as a throughput maximization, fairness optimization, or routing-scheduling co-design problem under half-duplex and bandwidth constraints. Although these studies provide valuable design insight, their evaluations are still predominantly based on analytical models or simulations, leaving implementation-level constraints largely abstracted away.

In parallel, O-RAN has enabled a programmable control framework through the near-RT RIC and xApps. Prior studies have shown that O-RAN can support closed-loop control, AI-driven decision making, and service-aware slicing on open experimental platforms, as illustrated by systems such as CoO-RAN, X5G, and ORANSlice [3]–[5]. These efforts are important because they demonstrate practical integration across O-RAN software components, OpenAirInterface-based stacks, and programmable control loops. However, most of them focus on general O-RAN experimentation, traffic steering, or slicing, rather than on the specific multi-hop bottlenecks that arise when IAB nodes must divide radio resources between upstream backhaul and downstream access.

Accordingly, the key gap in the literature is not the lack of IAB allocation algorithms, but the lack of empirical comparison on an open, programmable, multi-hop O-RAN IAB platform. Our work addresses this gap by combining OpenAirInterface, Docker Compose-based multi-host deployment, a BAP-less Linux routing design, and near-RT RIC/xApp control in a real SDR testbed. Rather than claiming that a single handcrafted strategy is universally optimal, we experimentally compare fixed-ratio PRB allocation, an adaptive xApp policy, and the default PF scheduler to reveal their throughput-latency trade-offs in practical IAB operation.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, we define the mathematical model for our multi-hop O-RAN environment and formulate the PRB allocation problem.

A. Network Architecture Model

Consider a tree-topology O-RAN IAB network composed of a 5G Core, a Donor Central Unit (CU), a Donor Distributed Unit (DU), a set of IAB nodes $\mathcal{K} = \{1, 2, \dots, K\}$, and a set of served User Equipments (UEs) $\mathcal{U} = \{1, 2, \dots, U\}$. Some IAB nodes operate as relay nodes, whereas others operate as access IAB nodes that serve downstream UEs. Let the total system bandwidth be divided into a set of Physical Resource Blocks $\mathcal{N}_{PRB} = \{1, 2, \dots, N\}$.

Because the IAB nodes share radio resources between upstream backhaul and downstream access transmission, each node must partition the available PRBs across these two functions. Under half-duplex and finite-buffer constraints, over-provisioning one side of the link pair can starve the other side and cause queue imbalance across hops.

B. Channel Model and Capacity

The Signal-to-Interference-plus-Noise Ratio (SINR) for a link between node i and node j on PRB n is given by:

$$\gamma_{i,j}^n = \frac{P_i^n |h_{i,j}^n|^2}{I_{i,j}^n + \sigma^2} \quad (1)$$

where P_i^n is the transmit power, $h_{i,j}^n$ is the channel gain, $I_{i,j}^n$ represents the interference from other transmitting nodes, and σ^2 is the additive white Gaussian noise (AWGN) variance.

The achievable data rate for the link over PRB n can be calculated using the Shannon capacity formula:

$$R_{i,j}^n = W_{PRB} \log_2(1 + \gamma_{i,j}^n) \quad (2)$$

where W_{PRB} is the bandwidth of a single PRB. The total throughput of a specific link is the sum of the rates over all allocated PRBs.

C. Problem Formulation

Our objective is to study how PRB allocation affects end-to-end service throughput and relay congestion while maintaining feasible flow transfer through the relay chain. Let $x_{i,j}^n \in \{0, 1\}$ denote whether PRB n is assigned to the link between nodes i and j . A reference optimization objective can be written as:

$$\max_{\mathbf{x}} \sum_{u \in \mathcal{U}} \sum_{n \in \mathcal{N}_{PRB}} R_{k,u}^n x_{k,u}^n \quad (3)$$

$$\text{s.t.} \quad \sum_{i,j} x_{i,j}^n \leq 1, \quad \forall n \in \mathcal{N}_{PRB} \quad (4)$$

$$R_{donor,k} \geq \sum_{u \in \mathcal{U}_k} R_{k,u}, \quad \forall k \in \mathcal{K} \quad (5)$$

Constraint (4) ensures that each PRB is assigned to at most one link simultaneously. Constraint (5) represents the flow conservation at the IAB node, ensuring that the backhaul capacity is sufficient to support the aggregated access traffic, thereby preventing bottleneck congestion.

In practice, our xApp is used as an experimental control mechanism rather than a guaranteed optimal solver. It can enforce predetermined PRB ratios and can also adapt the ratio according to the queue states of the donor-side backhaul and

IAB-side access links. This design is intentionally lightweight so that it can operate within the timing constraints of the near-RT RIC and the SDR-based OAI stack while enabling fair comparison among several allocation methods.

IV. PROPOSED IAB TESTBED ARCHITECTURE

To evaluate the PRB allocation schemes, we construct an experimental multi-hop IAB testbed using OpenAirInterface for the 5G protocol stack and USRPs for the RF front-end. The architecture can be explained from four technical dimensions: topology, infrastructure, forwarding design, and intelligent control.

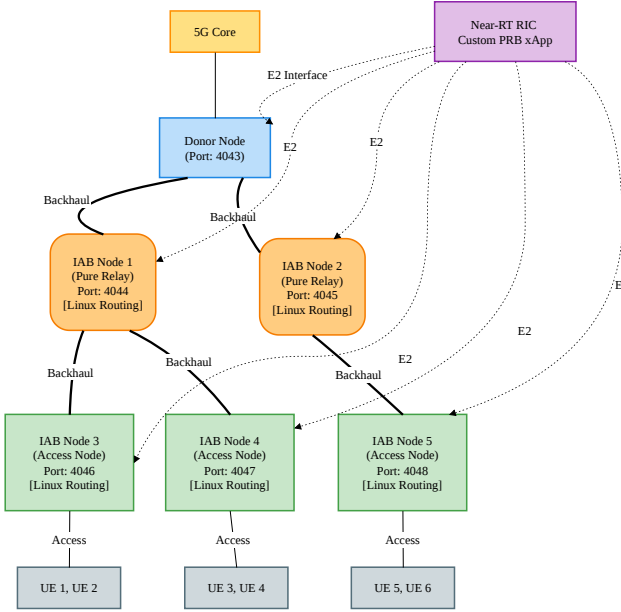


Fig. 1. The proposed tree-topology IAB architecture featuring Linux IP routing and xApp-based PRB allocation.

A. Topology Overview

Figure 1 illustrates the tree topology used in this work. The donor node connects the 5G Core to downstream IAB nodes. Intermediate nodes are organized into a relay layer that extends coverage hop by hop, while the leaf-side IAB nodes simultaneously maintain backhaul connectivity and provide access service to UEs. This separation of pure relay and access-capable relay roles makes it possible to study where bottlenecks emerge when traffic traverses multiple wireless hops.

B. Infrastructure and Software Stack

Each node is deployed on a high-performance Ubuntu 22.04 host equipped with a PREEMPT_RT low-latency kernel to preserve the timing determinism required by the OAI baseband stack. USRP B210 and N310 devices are used as the SDR front-end, and OAI provides the 5G NR protocol implementation for the donor, IAB, and UE functions. In our

implementation, the IAB-related functions are containerized and orchestrated with Docker Compose, which simplifies software management and improves experimental reproducibility across hosts while preserving access to the required radio and networking interfaces. This combination allows us to evaluate the PRB allocation strategy under realistic radio and host-processing constraints rather than relying only on software simulation.

From a deployment viewpoint, the testbed is split into a server-side Docker Compose stack and a client-side Docker Compose stack. The server-side stack hosts the O-RAN control components and core infrastructure, including FlexRIC, monitoring and control xApps, auxiliary inference services, MongoDB, the 5G Core, the donor node, and upper-layer IAB nodes. The client-side stack hosts the downstream access IAB nodes and the UE containers attached to them. This split deployment reflects the actual distribution of computation and radio functions across multiple PCs while keeping the full experiment manageable through declarative container orchestration.

The server/client split also reflects the physical distribution of the IAB hierarchy across multiple PCs.

C. BAP-less Implementation via Linux Routing

According to 3GPP specifications, IAB nodes rely on the BAP layer to manage routing and bearer mapping across multi-hop topologies. However, in open-source SDR deployments, a complete BAP implementation introduces additional encapsulation complexity, software overhead, and debugging difficulty.

To overcome this issue, we adopt a BAP-less architecture. In each IAB node, the IAB Mobile Termination (MT) for upstream connectivity and the IAB Distributed Unit (DU) for downstream service are co-located on the same physical host and deployed as containerized Docker Compose services. Instead of forwarding packets through a dedicated BAP layer, we use the native Linux kernel networking stack.

The OAI MT establishes a Protocol Data Unit (PDU) session toward the 5G Core through the donor node and creates a virtual interface such as `oaitun_ue1`. Meanwhile, the OAI DU maintains the tunnel and radio interfaces needed to serve downstream UEs. By enabling `net.ipv4.ip_forward=1`, configuring static routes, and applying `iptables` NAT rules when needed, packets can be transferred directly at the IP layer from the MT-facing interface to the DU-facing interface, even when the corresponding OAI functions are running inside containerized Docker Compose services with appropriately mapped network namespaces and interfaces. In this way, the node bypasses Layer-2 BAP encapsulation and performs lightweight multi-hop forwarding through Linux routing.

The container networks are also designed to match the experimental roles of the nodes. For example, the server-side stack uses a macvlan-based public network for inter-node signaling together with a bridge network for user-plane traffic toward the data network, whereas the client-side stack combines a macvlan segment for upstream IAB connectivity

and an internal bridge network for local MT, DU, and UE interaction. In several control-plane services, host networking is used to simplify access to SCTP- and E2-related interfaces and to avoid unnecessary address translation between the containerized O-RAN functions and the host system.

D. Near-RT RIC and xApp Integration

For the control plane, we integrate FlexRIC as the near-RT RIC platform. The E2 interface is established between FlexRIC and the MAC entities of the donor DU and IAB DUs, enabling centralized observation and control over the radio resource state. The custom xApp periodically retrieves Buffer Status Reports (BSRs) and Channel Quality Indicators (CQIs) through the E2SM-KPM service model, then computes an updated PRB split for backhaul and access traffic. The resulting control decisions are delivered through E2 control messages, allowing the MAC scheduler to react to traffic imbalance without modifying the overall O-RAN control architecture.

In the current implementation, the near-RT RIC and xApps are launched in the server-side Compose stack together with supporting database and inference services. This arrangement is useful because it co-locates the closed-loop control applications with the main orchestration environment, while the client-side Compose stack focuses on hosting the downstream IAB nodes and UEs that generate the radio measurements and traffic loads consumed by the control loop.

The xApps were deployed as separate containers and attached to the same Docker Compose-managed control environment as FlexRIC.

V. XAPP-BASED PRB ALLOCATION SCHEMES

We evaluate the system using different PRB allocation strategies, including fixed-ratio allocation, adaptive xApp control, and the default PF scheduler.

A. Scheme A: Fixed-Ratio Allocation

In the first class of methods, the total PRBs are partitioned using a fixed ratio between backhaul and access, such as 50/50, 30/70, or 10/90. These settings serve as controlled experimental points for observing how resource imbalance shifts the bottleneck across the multi-hop path. A balanced split may not always be best: under some traffic patterns it can under-provision backhaul, whereas under others it can leave access capacity underutilized.

B. Scheme B: Dynamic Traffic-Aware Allocation

To complement the fixed-ratio settings, we implemented a dynamic allocation algorithm. The xApp continually monitors the transmission buffers. If the buffer of the Donor DU (backhaul link) exceeds a predefined threshold while the IAB DU (access link) buffer is nearly empty, the xApp reallocates a larger percentage of PRBs to the backhaul link, and vice versa. The logic is summarized in Algorithm 1. This scheme is included as another comparison point; its purpose is to examine whether queue-aware control produces a different operating point from fixed ratios and from the default proportional fair (PF) scheduler.

C. Reference Scheduler: Default PF Scheduler

In addition to the xApp-controlled policies, we use the default proportional fair (PF) scheduler in OAI as a reference case. The PF scheduler is not expected to be dominated by every handcrafted PRB split, especially when channel-aware fairness and instantaneous radio conditions strongly influence scheduling decisions. Therefore, the purpose of our evaluation is to compare behaviors and trade-offs, not to claim that the xApp-based methods universally outperform PF in throughput.

Algorithm 1: Dynamic Traffic-Aware PRB Allocation

Require: BSR_{BH} (Backhaul Buffer), BSR_{AC} (Access Buffer)

Require: PRB_{total} , Threshold τ

- 1: Initialize $PRB_{BH} \leftarrow PRB_{total}/2$,
 $PRB_{AC} \leftarrow PRB_{total}/2$
- 2: **while** E2 connection is active **do**
- 3: Receive E2 Indication message
- 4: Update BSR_{BH} and BSR_{AC}
- 5: Calculate load ratio: $\rho = \frac{BSR_{BH}}{BSR_{BH} + BSR_{AC} + \epsilon}$
- 6: **if** $|\rho - 0.5| > \tau$ **then**
- 7: $PRB_{BH} \leftarrow \lfloor \rho \times PRB_{total} \rfloor$
- 8: $PRB_{AC} \leftarrow PRB_{total} - PRB_{BH}$
- 9: Encode E2 Control message with new allocation
- 10: Send E2 Control message to Donor DU and IAB DU
- 11: **end if**
- 12: Wait for next monitoring interval T_m
- 13: **end while**

VI. EXPERIMENTAL SETUP AND RESULTS

This section details the empirical evaluation of the testbed.

A. Setup Parameters

The testbed components are deployed across multiple high-performance PCs running Ubuntu 22.04 with a PREEMPT_RT low-latency kernel to ensure strict timing synchronization for the baseband processing. Table I lists the core parameters.

TABLE I
TESTBED EXPERIMENTAL PARAMETERS

Parameter	Value
Operating System	Ubuntu 22.04 (Low-latency Kernel)
Deployment Method	Docker Compose-based multi-host deployment
Deployment Split	Server-side and client-side stacks
CPU Governor	Performance Mode
SDR Hardware	USRP B210 / N310
Center Frequency	3.5 GHz (Band n78)
Channel Bandwidth	40 MHz
Subcarrier Spacing	30 kHz
Software Stack	OpenAirInterface (OAI) + Docker Compose
RIC Framework	FlexRIC
Total PRBs	106

B. Throughput Analysis

We generated UDP traffic using `iperf3` to stress the multi-hop path under increasing offered load. Under the same offered load, the 50/50 split achieved $X_{50/50}$ Mbps, the 30/70 split achieved $X_{30/70}$ Mbps, the 10/90 split achieved $X_{10/90}$ Mbps, the adaptive xApp achieved X_{dyn} Mbps, and the default PF scheduler achieved X_{PF} Mbps. These results indicate that the location of the bottleneck depends strongly on the selected PRB split. When more resources are reserved for backhaul, the relay can forward packets more aggressively, whereas a more access-oriented split may limit the end-to-end throughput once the backhaul becomes the dominant bottleneck.

Importantly, the purpose of this comparison is not to claim that a handcrafted policy always exceeds PF. Instead, the measured values reveal how each method behaves under the same traffic condition. If PF achieves the highest throughput, that outcome remains meaningful because it shows that static or queue-aware partitioning may trade peak throughput for other properties such as queue stability, resource isolation, or lower delay under a different operating region.

Fig. 2. Throughput comparison among fixed-ratio allocation, adaptive xApp control, and the default PF scheduler under varying network loads.

C. End-to-End Latency

Latency is a critical metric for 5G services. The BAP-less Linux routing inherently introduces some processing delay at the kernel level. However, poor PRB allocation exacerbates this by causing packets to wait in the MAC layer queues.

Fig. 3. Cumulative Distribution Function (CDF) of end-to-end latency for fixed-ratio allocation, adaptive xApp control, and the default PF scheduler.

Figure 3 illustrates the Cumulative Distribution Function (CDF) of the packet latency measured by ping RTT. Here again, the goal is to show how each method changes queueing behavior. A policy that does not maximize throughput may still reduce delay variation, whereas a policy with higher throughput may create a longer queue tail. Therefore, this section should report concrete values such as mean latency and 95th-percentile latency for each fixed ratio, including 50/50, 30/70, and 10/90, together with the adaptive xApp policy and the PF scheduler, then discuss the corresponding trade-offs.

Although PF delivered the highest throughput in this case, the 50/50 split showed a lower 95th-percentile latency, indicating a different throughput-latency trade-off. More specifically, the mean latency values of the 50/50 split, the 30/70 split, the 10/90 split, the adaptive xApp, and PF were $L_{50/50}$ ms, $L_{30/70}$ ms, $L_{10/90}$ ms, L_{dyn} ms, and L_{PF} ms, respectively, while the corresponding 95th-percentile latencies were $P_{50/50}$ ms, $P_{30/70}$ ms, $P_{10/90}$ ms, P_{dyn} ms, and P_{PF} ms. These results suggest that no single allocation rule is universally optimal across all traffic conditions.

D. System Resource Utilization

Operating OAI alongside a near-RT RIC requires substantial computational power. We monitored the CPU utilization of the IAB node during the experiments. Compared with the fixed-ratio settings, the adaptive xApp policy incurred only a negligible increase in CPU usage (less than 3%), confirming that the xApp control loop operates efficiently without starving the baseband processing threads.

VII. CONCLUSION

In this paper, we presented a practical multi-hop O-RAN IAB testbed and demonstrated that a BAP-less forwarding design based on Linux IP routing is feasible for SDR-based relay deployment. We further integrated a near-RT RIC and a custom xApp to implement multiple PRB allocation methods between backhaul and access links. The resulting platform enables direct experimental comparison among fixed-ratio configurations, an adaptive xApp policy, and the default PF scheduler in a real implementation environment. Our results show that these allocation methods exhibit different throughput-latency trade-offs, highlighting that the value of the proposed platform lies in empirically characterizing such behaviors rather than asserting that any single handcrafted policy consistently outperforms PF.

As future work, we plan to extend this framework toward hierarchical O-RAN control. In particular, a non-real-time RIC can host an rApp based on deep reinforcement learning (DRL) to learn long-term traffic patterns, topology states, and performance trends from historical observations. The learned policy can then be transferred from the non-RT RIC to the near-RT RIC, where an xApp executes the policy through near-real-time PRB control over E2-connected nodes. Such collaboration between the non-RT RIC and near-RT RIC is expected to better align with the O-RAN architecture and to support more adaptive throughput-latency trade-offs under diverse traffic conditions.

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